

1) CRD/Completely Randomized Design (CRD)

- * It is the simplest experimental design.
- * Treatments are assigned to experimental units completely at random, so every unit has an equal chance of receiving any treatment.
- * Suitable when experimental units are homogeneous (same soil, environment, etc.).

Fundamental Principles Applied.

Randomization: Assign treatments randomly to avoid bias.

Replication: Repeat each treatment several times to estimate experimental error.

No blocking: Not required since all units are similar.

(c) layout

Treatment Replication 1 Replication 2 Replication 3

T1

T2

T3

Sum.

(d) ANOVA.

(e) when CRD is preferred:

(i) when units are uniform & few in number.

(ii) In laboratory or greenhouse experiments.

(f) limitation:

(i) Not suitable if there is heterogeneity among experimental units.

(ii) Random variation increases error if conditions differ.

2) Randomized Block Design.

* In RBD, experimental units are divided into blocks, so that units within a block are homogeneous.

* Treatments are randomly assigned within each block.

* Controls variability due to known sources (like soil fertility).

(b) Fundamental Principles

(i) Randomization: Treatments randomly assigned in each block.

(ii) Replication: Each treatment appears once per block.

(iii) Blocking: Reduces variability among experimental units.

(c) layout

Block T1 T2 T3

B1

B2

B3

(d) Analysis

(e) Advantage over CRD

* Blocking reduces experimental error by removing variation among blocks.

* Gives more precise treatment comparisons.

(f) Limitation:

(i) not suitable when many treatments make block size large.

(ii) Assumes no interaction b/w treatments & block.

* 5D (Latin Square Design)

* LSNs used when there are 2 sources of variation (rows & columns) in addition to treatments.

* Every treatment appears once in each row & column.

* Suitable for experiments where 2 nuisance factors (e.g. soil fertility gradient & irrigation) affect results.

(b) Principles Applied

(i) Randomization: Treatments randomly assigned to rows & columns.

(ii) Replication: Each treatment occurs once in each row & column.

(iii) Blocking: Controlled in 2 directions (row & column).

(c) Layout Row/Column E1 E2 E3 E4

	R1	R2	R3	R4
C1	T1	T2	T3	T4
C2	T2	T3	T4	T1
C3	T3	T4	T1	T2
C4	T4	T1	T2	T3

(d) Analysis

sum

(e) ^{Adv:} (i) controls 2 sources of variation simultaneously.

(ii) Reduces experimental error

further than CRD or RPB.

Limitation

(i) no of treatments = no of rows = no of columns

(ii) missing data or unequal replication can complicate analysis.

Unit - 5

Sampling & Population in Statistics

CRD - 1 way sum

	Replications		
	R1	R2	R3
T1	1.5	1.8	1.6
T2	4.7	4.4	5.3
T3	5.7	5.8	5.3
T4	3.7	3.2	2.9
T5	6.6	6.3	6
T6	5.8	6.1	5.5

(i) Find $N = 6 \times 3 = 18$

(ii) $T = 82.2$

(iii) $C.F = \frac{T^2}{N} = \frac{(82.2)^2}{18} = \frac{6756.84}{18}$

$= 375.38$

(iv) $TSS = [\sum R_1^2 + \sum R_2^2 + \sum R_3^2] - C.F$

$= [147.72 + 143.38 + 133.4] - 375.38$

$= 49.12$

H_0 : There is no significant difference among the treatments

H_1 : There is a significant difference among the treatments.

~~H_0 is accepted~~

$$(v) SST = \sum_{n=3} \left[\frac{(\sum R_1)^2}{n} + \frac{(\sum R_2)^2}{n} + \frac{(\sum R_3)^2}{n} + \frac{(\sum R_4)^2}{n} + \frac{(\sum R_5)^2}{n} + \frac{(\sum R_6)^2}{n} \right] - C.F$$

$$= [8.003 + 69.12 + 94.08 + 32.01 + 119.07 + 100.92] - 375.38$$

$$= 47.82$$

$$(vi) SSE = TSS - SST = 49.12 - 47.82 = 1.3$$

7) Anova table

Source of Variation	Sum of Squares	D.O.f	mean sum of squares	Variance Ratio	Table Value (0.05)
Treatment	$SST = 47.82$	$R-1 = 6-1 = 5$	$MSE = SST/C-1 = 47.82/5 = 9.564$	$F_C = MSC/MSE = 88.51$	$F(2, 15) = 3.682$
Error	$SSE = 1.3$	$N-C = 18-6 = 12$	$MSE = 1.3/12 = 0.1083$	$= 9.56/0.1083$	
Total	$TSS = 49.12$	$N-1 = 18-1 = 17$		$F_C = 88.51$	

Cal > Table

Value

Value

is rejected.

RCBD (Randomized Complete Block Design) or Two-way Anova classification.
 * blocks made.

* Can be effectively analyzed through Anova.

3 sources of variation in RCBD

(i) Treatment → variance of interest.

(ii) Block → unavoidable variation.

(iii) Experimental error

* Fresh weight of plants under different doses of Fertilizer.

Treatments	Blocks			Total			
	I (x_1)	II (x_2)	III (x_3)		x_1^2	x_2^2	x_3^2
A (y_1)	30.85	38.01	35.10	103.96	951.72	1444.56	1232.01
B (y_2)	30.24	28.43	35.93	94.60	914.45	808.26	1290.96
C (y_3)	30.94	31.64	34.95	97.53	957.28	1001.08	1221.50
D (y_4)	29.89	29.12	36.75	95.76	893.41	847.97	1350.56
E (y_5)	21.52	24.07	20.76	66.35	463.11	579.36	430.97
F (y_6)	25.38	32.14	32.19	89.71	644.14	1032.97	1036.19
G (y_7)	28.89	19.66	26.92	69.47	523.95	386.51	724.68
H (y_8)	29.44	24.15	37.99	92.38	866.71	622.50	1433.24
Total	$\sum x_1$ = 221.15	$\sum x_2$ = 228.02	$\sum x_3$ = 260.59	709.76	$\sum x_1^2$ = 6214.77	$\sum x_2^2$ = 6723.41	$\sum x_3^2$ = 8639.11

(i). H_0 : The μ s no significant difference b/w blocks & treatments.

H_1 : There is a significant difference b/w blocks & treatments.

(ii) $N = 24$ (i) $T = 709.76$ (iii) $C.F = \frac{T^2}{N} = 20,989.96$

(iv) $TSS = [\sum x_1^2 + \sum x_2^2 + \sum x_3^2] - C.F = [6214.77 + 6723.41 + 8639.11] - 20,989.96$
 $= 21,577.29 - 20,989.96$

$TSS = 587.33$ b/w

$$\begin{aligned}
 (v) SST &= \left[\frac{(103.96)^2}{3} + \frac{(94.60)^2}{3} + \frac{(97.53)^2}{3} + \frac{(95.76)^2}{3} + \frac{(66.35)^2}{3} \right. \\
 &\quad \left. + \frac{(89.71)^2}{3} + \frac{(69.47)^2}{3} + \frac{(72.38)^2}{3} \right] - C.F \\
 &= [3602.56 + 2983.05 + 3170.70 + 3056.65 + 1467.4 + 2682.02 \\
 &\quad + 1608.69 + 2844.67] - 20,989.96 \\
 &= 21,416.35 - 20,989.96 \\
 &= 426.39
 \end{aligned}$$

$$\begin{aligned}
 (vi) SSB &= \left[\frac{(\sum x_1)^2}{8} + \frac{(\sum x_2)^2}{8} + \frac{(\sum x_3)^2}{8} \right] - C.F \\
 &= \left[\frac{(222.15)^2}{8} + \frac{(228.02)^2}{8} + \frac{(260.59)^2}{8} \right] - 20,989.96 \\
 &= [6113.41 + 6499.14 + 8488.39] - 20,989.96 \\
 &= 21,100.94 - 20,989.96 \\
 &= 110.98
 \end{aligned}$$

$$\begin{aligned}
 (vii) SSE &= TSS - SST - SSB = 678.42 - 426.39 - 110.98 \\
 &= 141.05
 \end{aligned}$$

viii) Anova table

Source of Variation	Sum of squares	D. of	mean Sum of squares	Variance ratio	Table Value (5%)
Treatment	$SST = 426.39$	$r-1 = 8-1 = 7$	$MSA = \frac{426.39}{7} = 60.91$	$F_c = \frac{60.91}{10.07} = 6.048$	$F(7, 14) = 2.76$
Blocks	$SSB = 110.98$	$s-1 = 3-1 = 2$	$MSB = \frac{110.98}{2} = 55.49$	$F_c = \frac{55.49}{10.07} = 5.51$	$F(2, 12) = 1$
Error	$SSE = 141.05$	$(r-1) \times (s-1) = 7 \times 2 = 14$	$MSE = \frac{141.05}{14} = 10.07$		
Total	$TSS = 678.42$	$rc-1 = 23$			

For treatment
Calculate
value > Table
value

H_0 is rejected

For block
Calculate
value > Table
value

H_0 is rejected.

2) RBD another sum.

Workers	machine type			
	A	B	C	D
1	44	38	47	36
2	46	40	52	43
3	34	36	44	32
4	43	38	46	33
5	38	42	49	39

1) CRD Completely randomized Design (CRD)

* It is the simplest experimental design.

It is assigned to experimental units

Latin square design (LSD Sum).

A farmer wishes to test the effects of 4 different fertilizers A, B, C, D on the yield of wheat. In order to eliminate sources of error due to variability in soil fertility, he uses the fertilizers in a Latin square arrangement as indicated with following table, where the numbers indicate yields in bushels per unit area. Perform an analysis of variance to determine if there is a significant difference b/w the fertilizers at $\alpha = 0.05$ level of significance.

	X1	X2	X3	X4
Y1	A 18	C 21	D 25	B 11
Y2	D 22	B 12	A 15	C 19
Y3	B 15	A 20	C 23	D 24
Y4	C 22	D 21	B 10	A 17

$$(18)^2 + (22)^2$$

1) H_0 : There is no difference b/w rows, columns and treatments.

H_1 : There is a difference b/w the rows, columns & treatments.

2) $N = 16$

$$3) T = 18 + 21 + 25 + 11 + 22 + 12 + 15 + 19 + 15 + 20 + 23 + 24 + 22 + 21 + 10 + 17$$

$$= 295$$

$$4) C.F = \frac{T^2}{N} = \frac{(295)^2}{16} = 5439.0625$$

$$5) TSS = \left[\sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 \right] - C.F$$

$$= [1517 + 1426 + 1479 + 1347] - 5439.0625$$

$$= 5769 - 5439.0625$$

$$= 329.93$$

$$\begin{aligned}
 (5) \text{ SSR} &= \left[\frac{\sum x_1^2}{n} + \frac{\sum y_2^2}{n} + \frac{\sum y_3^2}{n} + \frac{\sum y_4^2}{n} \right] - \text{C.F} \\
 &= \left[\frac{(75)^2}{4} + \frac{(68)^2}{4} + \frac{(82)^2}{4} + \frac{(70)^2}{4} \right] - 5439.0625 \\
 &= \frac{1}{4} [21,873] - 5439.0625 \\
 &= 5468.25 - 5439.0625
 \end{aligned}$$

$$\text{SSR} = 29.1875$$

$$\begin{aligned}
 (6) \text{ SSC} &= \left[\frac{(\sum x_1)^2}{n} + \frac{(\sum x_2)^2}{n} + \frac{(\sum x_3)^2}{n} + \frac{(\sum x_4)^2}{n} \right] - \text{C.F} \\
 &= \frac{1}{4} [(77)^2 + (74)^2 + (73)^2 + (71)^2] - 5439.0625 \\
 &= 5443.75 - 5439.0625 \\
 &= 4.6875
 \end{aligned}$$

(7) Treatment (rowwise) arrange the data

					Total
A	18	15	20	17	70
B	11	12	15	10	48
C	21	19	23	22	85
D	25	22	24	21	92

$$\begin{aligned}
 \text{SST} &= \left[\frac{(70)^2}{4} + \frac{(48)^2}{4} + \frac{(85)^2}{4} + \frac{(92)^2}{4} \right] - 5439.0625 \\
 &= 5723.25 - 5439.0625 = 284.1875
 \end{aligned}$$

$$\textcircled{8} \text{ SSE} = \text{TSS} - \text{SSC} - \text{SSR} - \text{SST}$$

$$= 329.93 - 40.6875 - 29.1875 - 284.1875$$

$$= 11.8675$$

⑨ ANOVA table

df (error value, df)

Source of Variation	Sum of squares	D. of	mean sum of squares	Variance ratio	Table value 0.05%
B/w Columns	SSC = 40.6875	$n-1$ = 3	$\text{MSC} = \frac{\text{SSC}}{n-1}$ = $\frac{40.6875}{3}$ = 13.5625	$F_C = \frac{\text{MSE}}{\text{MSC}}$ = $\frac{1.9779}{13.5625}$ = 1.2658	$F_C(6, 3)$ = 8.94
B/w Rows	SSR = 29.1875	$R-1$ = 3	$\text{MSR} = \frac{\text{SSR}}{R-1}$ = $\frac{29.1875}{3}$ = 9.7291	$F_R = \frac{\text{MSR}}{\text{MSE}}$ = $\frac{9.7291}{1.9779}$ = 4.9189	$F_C(3, 6)$ = 4.76
B/w treatments	SST = 284.1875	$n-1$ = 3	$\text{MST} = \frac{\text{SST}}{n-1}$ = $\frac{284.1875}{3}$ = 94.7291	$F_T = \frac{94.7291}{1.9779}$ = 47.893	$F_C(3, 6)$ = 4.76
Error	SSE = 11.8675	$n-1 \times n-2$ = 3 x 2 = 6	$\text{MSE} = \frac{\text{SSE}}{n-1 \times n-2}$ = $\frac{11.8675}{6}$ = 1.9779		

MSE is high (6, 3)

(Treat on df)

⑩ (i) For columns

Calculated $<$ Table F_C Value
 F_C Value

H_0 is accepted.

(ii) For rows

Calculated $>$ Table F_R Value
 F_R Value

H_0 is rejected.

(iii) For treatments
Calculated $>$ Table F_T Value
 F_T Value
 H_0 is rejected

A variable trial was conducted on wheat with 4 varieties in a LSD. The plan of the experiment and per plot yield are given below: $\alpha = 5\%$

C25 B23 A20 D20
A19 D19 C21 B18
B19 A14 D17 C20
D17 C20 B21 A15